Stable LatentMoE

Half-width routed experts with a normalized, bounded latent path

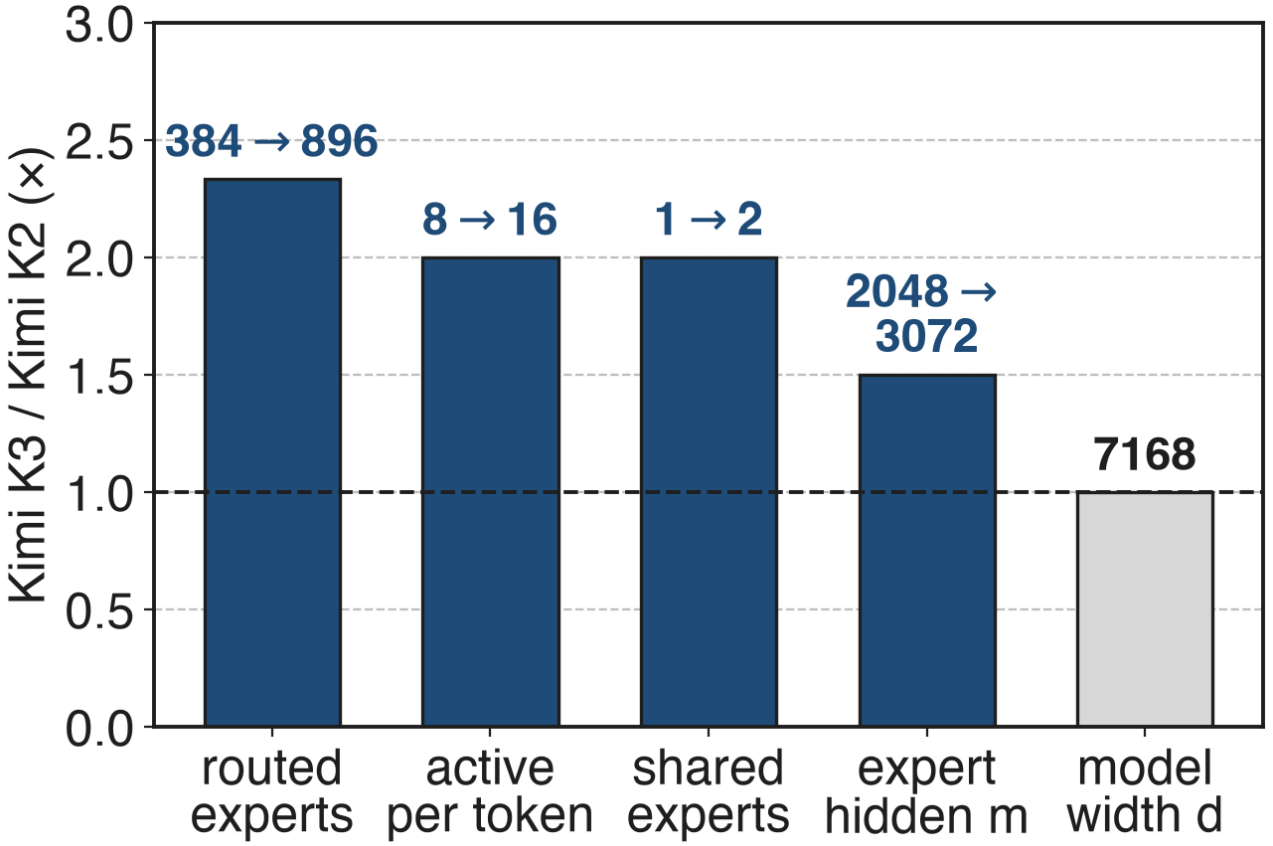
Moonshot AI · Kimi K3

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

MoE & Routing · arxiv.org/abs/2607.24653

K3 doubles K2's active experts at equal width

- Kimi K3: **2.8T** params, **104B** active
- **16** of **896** routed experts: sparsity **56**
- Model width d stays **7,168**



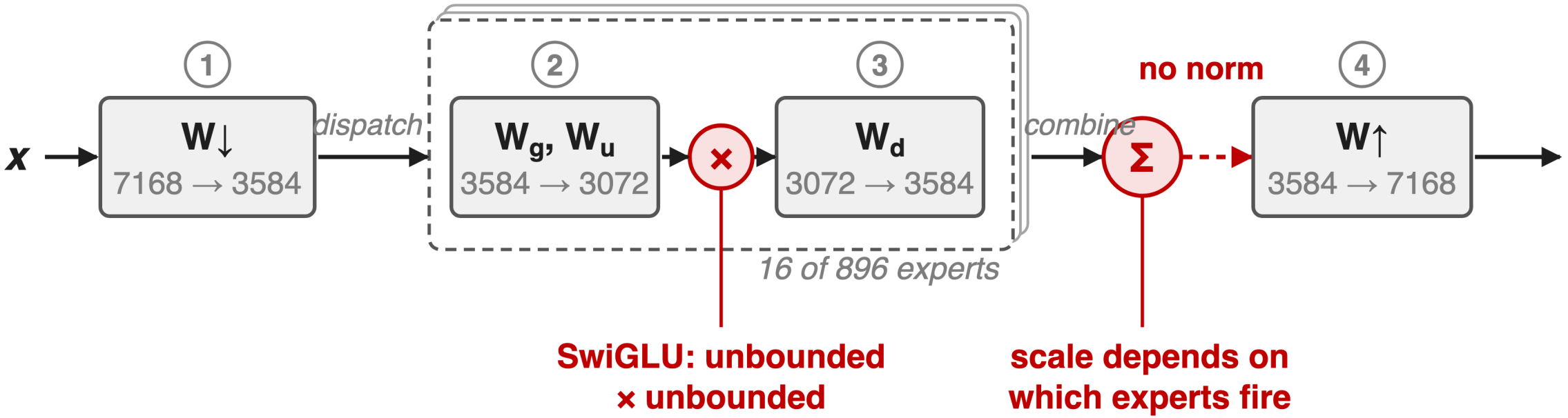
Full-width experts would double the all-to-all

2×

dispatch + combine volume per token if K3's 16 experts each took the full 7,168-wide token, vs K2's 8 (analytic)

K3 Table 1 (arXiv 2607.24653); volume ∝ top-k × routed width, LatentMoE §2.2 (arXiv 2601.18089)

Plain LatentMoE stacks four matmuls, no norm



At 896 experts and 2.8T parameters this chain produced exploding activations

Insight: narrow is safe; scale is not

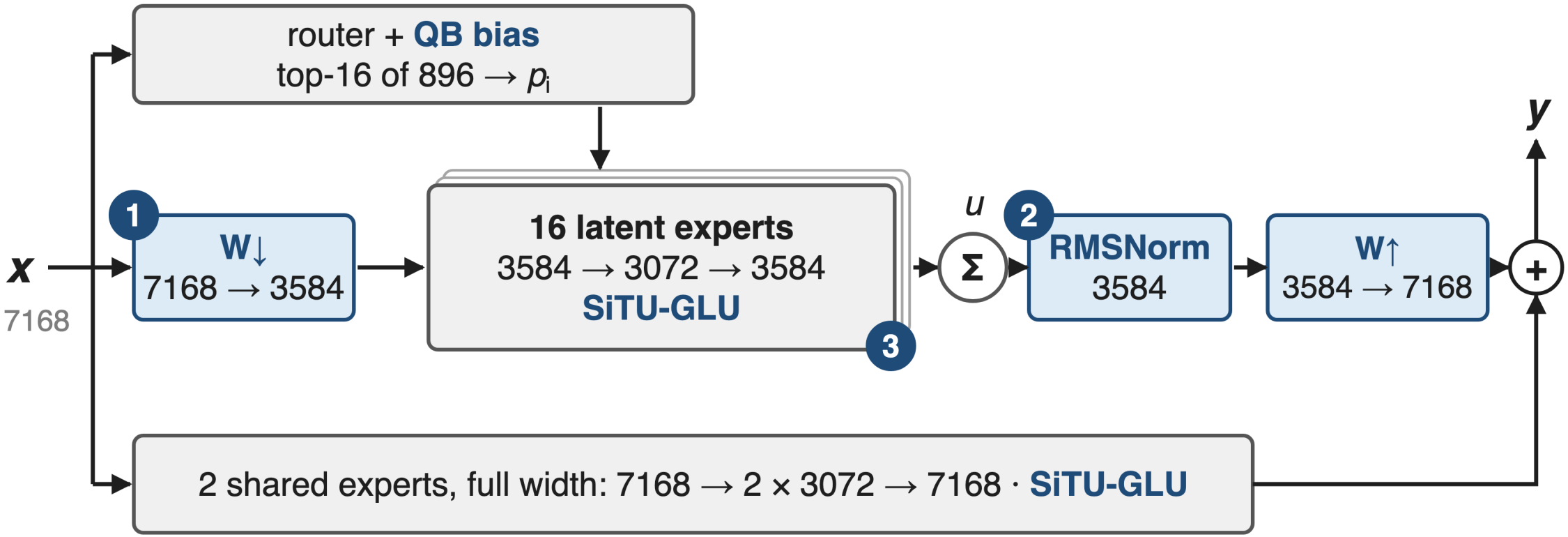
- LatentMoE: quality holds up to **4×** narrower experts
- K3 goes **2×** narrower and doubles top-k
- Blow-ups come from scale: routed sum, GLU product

Insight: narrow is safe; scale is not

- LatentMoE: quality holds up to **4×** narrower experts
- K3 goes **2×** narrower and doubles top-k
- Blow-ups come from scale: routed sum, GLU product

Keep the narrow path; bound its scale where it can grow

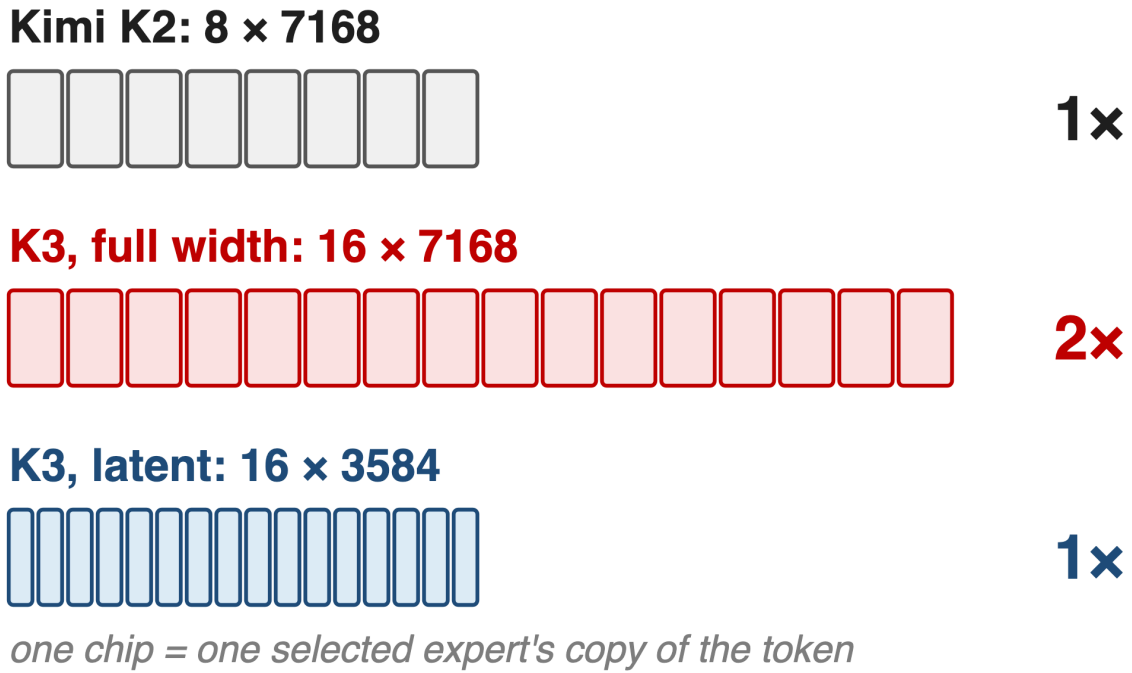
Stable LatentMoE: narrow, normalized, bounded



❶ latent W↓ / W↑ (from LatentMoE) · ❷ RMSNorm on the weighted sum · ❸ SiTU-GLU in every FFN · balancing: Quantile Balancing (deck 56)

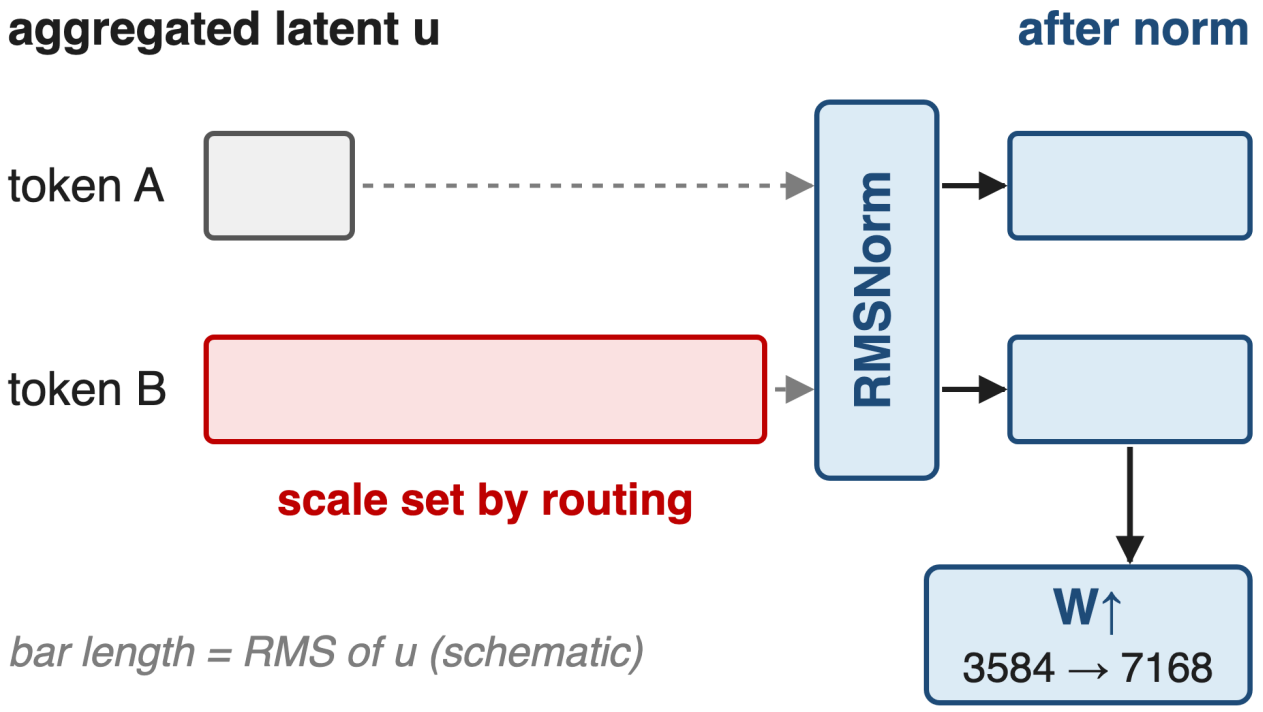
❶ Sixteen half-width experts cost K2's bytes

- Shared **W↓** to 3,584 before dispatch
- Experts: 3,584 → 3,072 → 3,584
- Shared **W↑** after combine



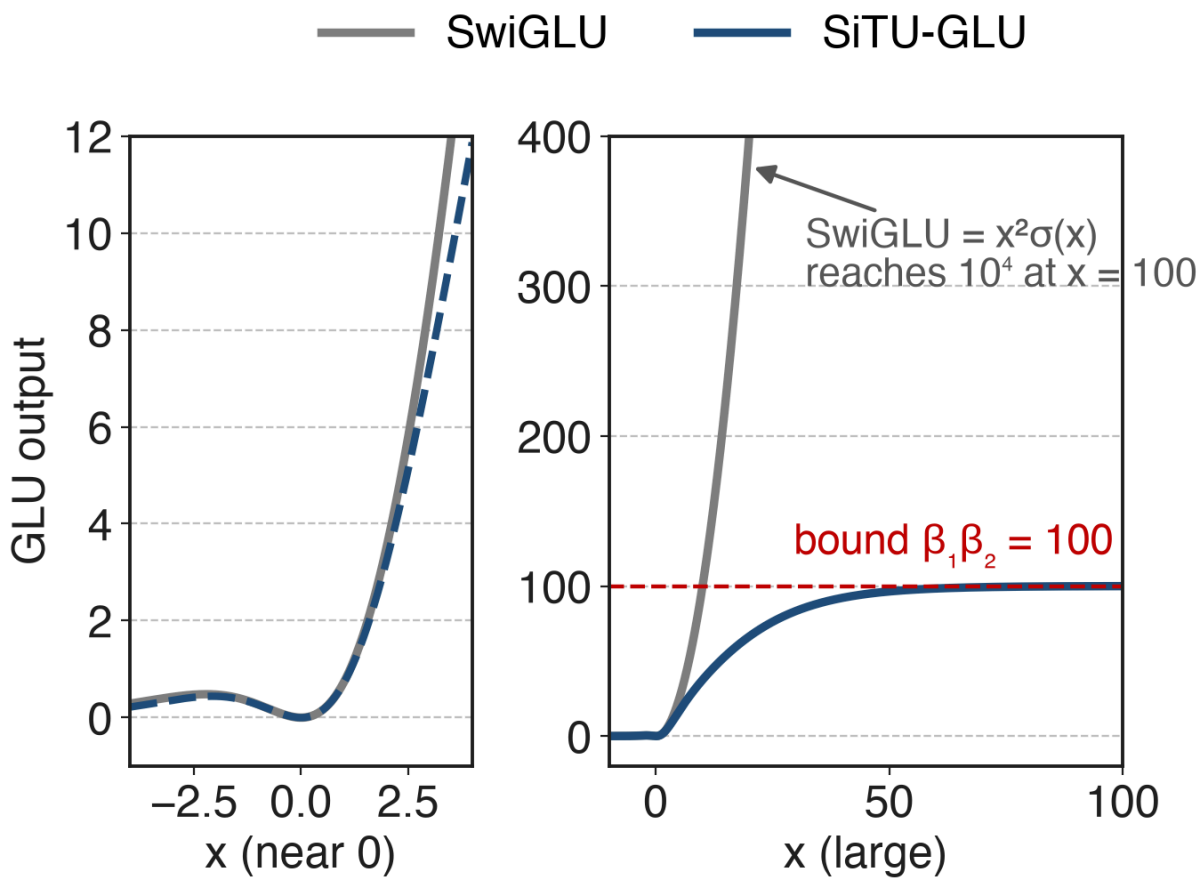
❷ RMSNorm fixes the latent's scale before W↑

- u: weighted sum of **16** expert outputs
- Its scale depends on the routing
- Normalize, then W↑, then add shared



❸ SiTU-GLU caps both GLU factors smoothly

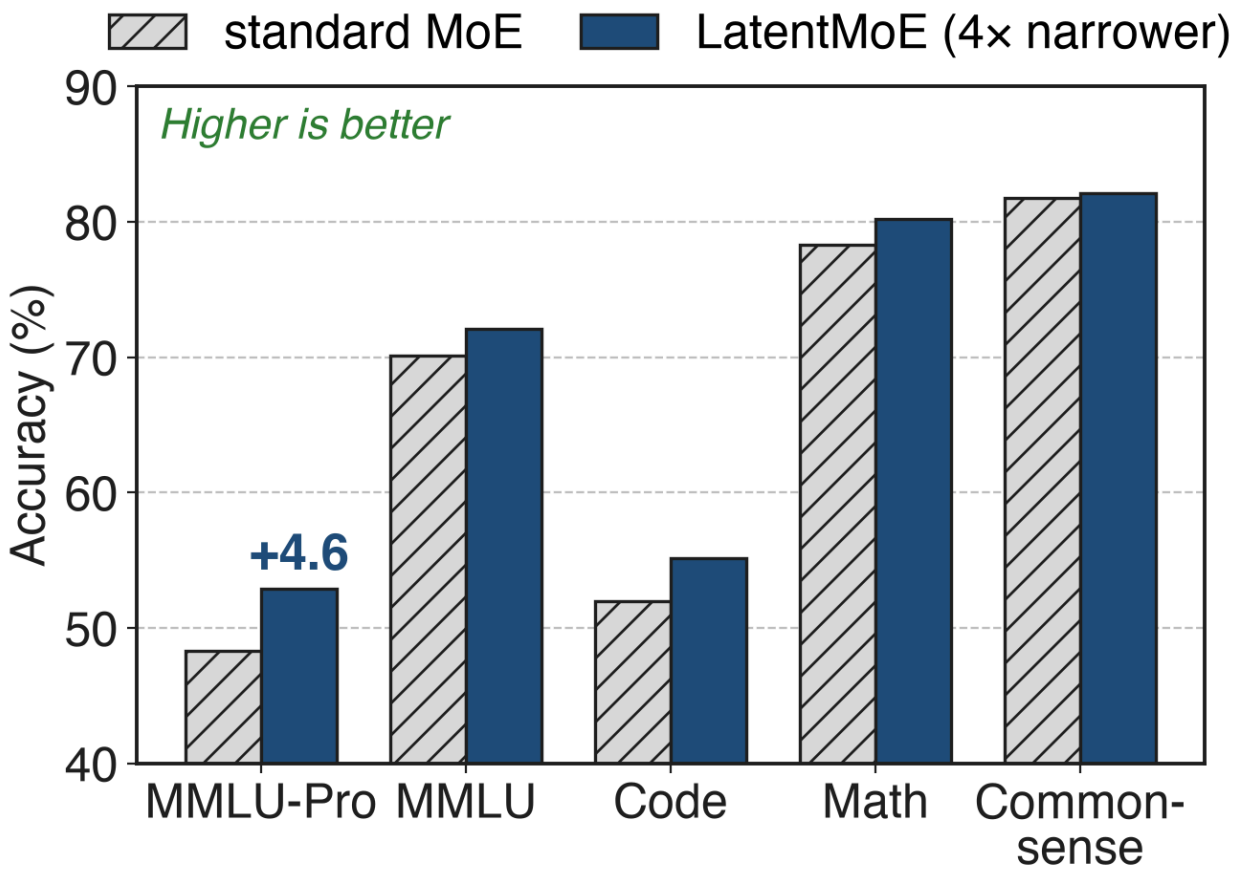
- Gate: 4·tanh(g/4)·σ(g)
- Up: 25·tanh(u/25)
- ≈ SwiGLU near 0; output ≤ **100**



Analytic: both branches fed the same x, as in K3 Fig. 4

Latent experts add +4.6 MMLU-Pro at equal cost

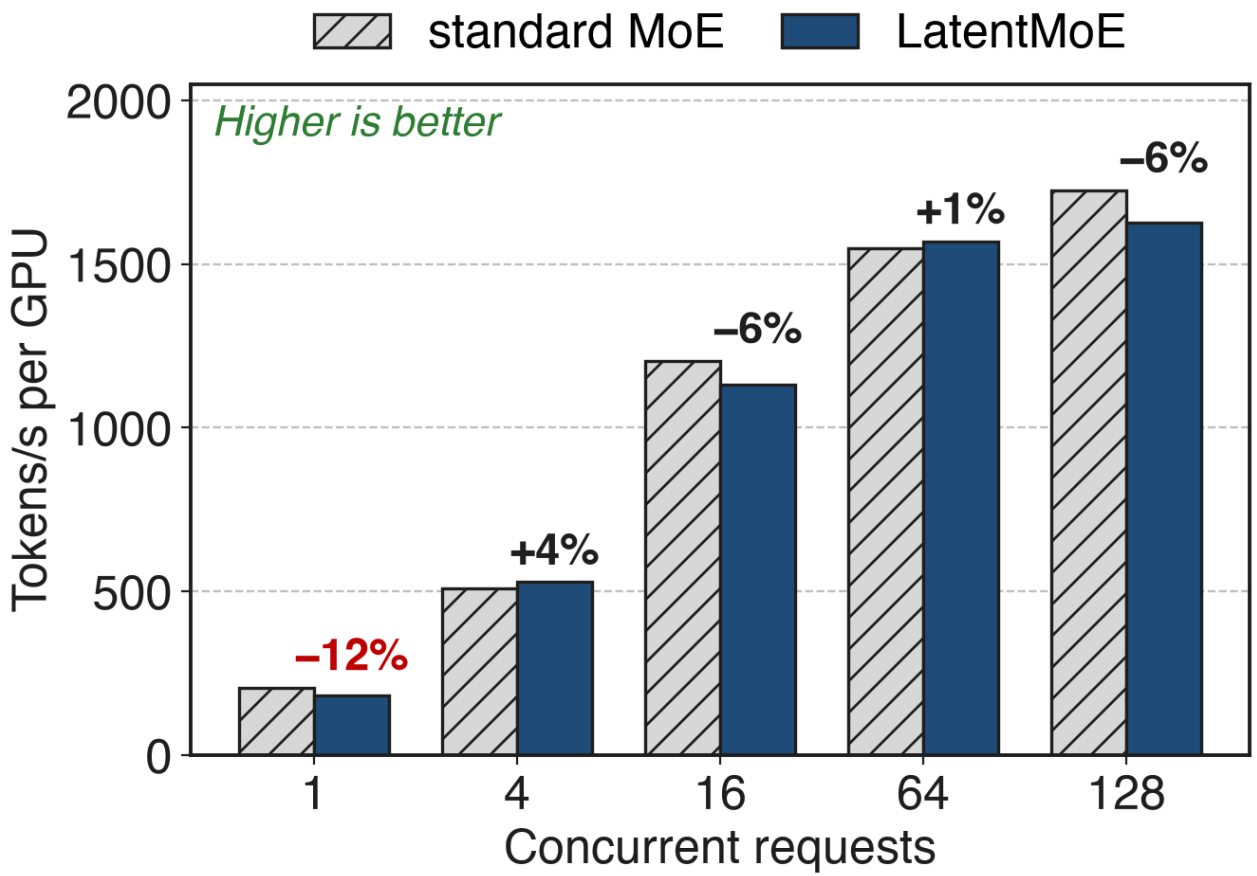
- Same active (**8.0B**), total (**73B**) params
- 1T tokens, hybrid Mamba-attention MoE
- Plain LatentMoE, no K3 stabilizers



NVIDIA LatentMoE paper (arXiv 2601.18089), Table 4

The latent detour costs up to 12% throughput

- **–12%** at concurrency 1
- Within **6%** at 4–128
- Extra W↓ / W↑ GEMMs, smaller expert GEMMs



Two H100, vLLM, FP8, hybrid 73B MoE (LatentMoE Table 5)

K3 trains 16-of-896 routing at 2.8T scale

2.5×

K3's overall scaling-efficiency gain over Kimi K2: KDA, AttnRes, Stable LatentMoE and new recipes together

Kimi K3 report (arXiv 2607.24653), abstract, §3.2 and Fig. 7; no isolated Stable LatentMoE ablation is published

The price: two extra GEMMs and no ablation

- **Two extra GEMMs**: W↓, W↑, 51M params per layer
- K3 fuses W↓ into the router GEMM
- More, smaller expert GEMMs strain kernels
- Width floor: quality drops past **4×**
- **No K3 ablation** of RMSNorm or SiTU
- Needs exact balance: QB (deck 56)

Summary

- 16 full-width experts: **2×** K2's all-to-all bytes
- **Insight**: narrow experts work; bound the scale instead
- Latent W↓/W↑, RMSNorm on the sum, SiTU-GLU (≤ **100**)
- **+4.6** MMLU-Pro at equal cost; K3 runs 16 of **896** at **2.8T**

https://arxiv.org/abs/2607.24653